

Abstract :Smart Materials in Civil Engineering

Smart materials are advanced materials that can sense and respond to changes in their environment such as stress, temperature, moisture, and damage. In civil engineering, these materials are used to improve the safety, durability, and performance of structures. Examples of smart materials include self-healing concrete, shape memory alloys, piezoelectric materials, Fiber optic sensors, and phase change materials. They are applied in smart buildings, bridges, roads, and earthquake-resistant structures. Smart materials help in reducing maintenance costs, increasing the lifespan of structures, and supporting sustainable construction practices. With the growth of modern technology, smart materials are expected to play an important role in the future development of intelligent and resilient civil engineering infrastructure.